Two Contact Valuations and a Rational Function

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Abstract

No fixed nonzero rational function of the input and slope can have odd valuation whenever either of two independent contact values does. We prove this statement on the exact eighth-power scale used in the thirteen-variable construction under weak *abc*. The counterexamples have positive slopes that are sums of two integer squares, nonintegral rational inputs, and an input residual that is a unit at the prime in question. The result includes arbitrary rational combinations of the two contact values, with explicit dependence on the input or scale allowed. It concerns a uniform local parity implication and gives no lower bound for arbitrary Diophantine definitions.

1 The contact function and the statement

For a prime p, write v_p for the valuation on $\mathbb{Q}_p^\times$ normalized by $v_p(p) = 1$. Put

$$R(t) = \frac{t(t+1)(2t+1)(3t+1)(4t+1)(5t+2)}{120(t-1)^8}. \tag{1}$$

Its six simple zeros are

$$0, \quad -1, \quad -\frac{1}{2}, \quad -\frac{1}{3}, \quad -\frac{1}{4}, \quad -\frac{2}{5}.$$

Fix an odd integer $B \geq 3$, an integer $n \neq 0$, and a rational number $T > 6$. For $X \in \mathbb{Q}$, define

$$\beta(X) = 2X^2 + B, \qquad a(X) = \beta(X)^8 + T^8. \tag{2}$$

In the application in [1], one takes $T = T(n) = \mathrm{Tr}_{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}\, x([n]P)$. The argument below uses only that this is a fixed rational number greater than 6; it uses no *abc* inequality.

Theorem 1.1. *Let $\Psi \in \mathbb{Q}(X, L)$ be nonzero, and let S be any finite set of primes. There are an integer $C \geq 1$, a nonzero vector $\epsilon \in \{0,1\}^2$, and infinitely many primes $p \notin S$ with the following property.*

For every pair of positive integers μ, ν satisfying

$$\nu > C\mu, \qquad (\mu, \nu) \equiv \epsilon \pmod 2,$$

there exist rational numbers X, λ such that

$$X \in \tfrac{1}{2}\mathbb{Z} \setminus \mathbb{Z}, \qquad \lambda = s^2 + t^2 > 1 \quad (s, t \in \mathbb{Z}), \tag{3}$$

$$v_p(R(\lambda)) = \mu, \qquad v_p(R(a(X)\lambda)) = \nu, \tag{4}$$

$$\Psi(X, \lambda) \neq 0 \quad \textit{is defined}, \qquad v_p(\Psi(X, \lambda)) \in 2\mathbb{Z}, \qquad v_p(\beta(X) - n) = 0. \tag{5}$$

For each such p, both depths can be made arbitrarily large. After fixing the depths, λ and $|X|$ can also be made arbitrarily large.

In particular, the implication

$$(v_p(R(\lambda)) > 0 \text{ odd or } v_p(R(a(X)\lambda)) > 0 \text{ odd}) \implies v_p(\Psi(X,\lambda)) \text{ odd}$$

cannot hold uniformly on this parameter domain, even after discarding any prescribed finite set of primes. Since $X = q/2$ with q odd in the theorem, $\beta(X) = q^2/2 + B$ is itself nonintegral. The theorem allows Ψ to depend explicitly on $\beta(X)$ or $a(X)$, since both are rational functions of X.

2 A valuation lemma

The elementary obstruction is slightly more general than the case of two contact values.

Lemma 2.1. *Let $1 \le m < d$, and let $\Phi_1, \dots, \Phi_m \in \mathbb{Q}(Z_1, \dots, Z_d)$ be nonzero rational functions. There is a finite set of primes S_0 and a sequence*

$$w_N = (w_{N,1}, \dots, w_{N,d}) \in \mathbb{Z}^d_{>0}$$

such that every coordinate tends to infinity, at least one coordinate of every w_N is odd, and the following holds. For every $p \notin S_0$ and every $z_1, \dots, z_d \in \mathbb{Q}_p^\times$ with $v_p(z_i) = w_{N,i}$, all the $\Phi_j(z_1, \dots, z_d)$ are defined and nonzero and have even valuation.

Proof. Write $\Phi_j = P_j/Q_j$ with nonzero polynomials $P_j, Q_j \in \mathbb{Z}[Z_1, \dots, Z_d]$. Exclude the finitely many primes dividing a nonzero coefficient of any of these polynomials.

For each polynomial, compare the linear forms $\langle e, w\rangle$ belonging to its distinct exponent vectors e. The equalities between these forms lie on finitely many proper hyperplanes. Choose w in the positive orthant off all these hyperplanes. There is an open cone C containing w on which each P_j and each Q_j has a unique monomial of smallest weight. Let their exponent vectors be u_j and v_j, respectively. If the valuation vector of z lies in C, the unique term of smallest valuation cannot cancel. Consequently

$$v_p(\Phi_j(z)) = \langle u_j - v_j, v_p(z)\rangle. \tag{6}$$

This also proves that both evaluated polynomials are nonzero.

The matrix with rows $u_j - v_j$, reduced modulo 2, has $m < d$ rows. Its kernel over $\mathbb{F}_2$ contains a nonzero vector $\epsilon \in \{0,1\}^d$. Choose a positive integer vector $v \in C$, and set

$$w_N = 2Nv + \epsilon.$$

For all sufficiently large N, this vector lies in C. Every coordinate tends to infinity, its parity vector is the fixed nonzero vector ϵ, and all the expressions in (6) are even. Discarding the finitely many initial values of N proves the statement. □

The lemma concerns valuations of rational functions of d independent values. Its linear-algebra statement is not a variable-count lower bound for arbitrary systems of polynomial equations.

3 Simple roots of the prescribed scale

We give the elementary prime-divisor argument needed to retain the exact scale (2).

Lemma 3.1. *Let $G \in \mathbb{Z}[X]$ be nonconstant with $G(0) \neq 0$. There are infinitely many primes dividing nonzero values $G(k)$, $k \in \mathbb{Z}$. If G is separable over $\mathbb{Q}$, then outside any finite set there are infinitely many primes at which G has a simple root modulo p.*

Proof. Put $c = G(0)$. Given a finite set of primes, let M be their product, with $M = 1$ for the empty set. The polynomial

$$H(Y) = G(cMY)/c$$

has integer coefficients, is nonconstant, has constant term 1, and is congruent to 1 modulo every prime dividing M. Choose an integer k with $|H(k)| > 1$. A prime divisor p of $H(k)$ lies outside the specified set and divides the nonzero integer $G(cMk)$. This proves infinitude.

If G is separable, enlarge the excluded set by the primes dividing its leading coefficient and its nonzero discriminant. A root of G modulo any remaining prime is simple. □

Lemma 3.2. *The polynomial $a(X) - 1$ is separable over $\mathbb{Q}$. Fix any one of its roots α, and put $K = \mathbb{Q}(\alpha)$. Outside any prescribed finite set, there are infinitely many primes p admitting an embedding $\iota : K \hookrightarrow \mathbb{Q}_p$ such that, with $\xi = \iota(\alpha)$,*

$$\xi \in \mathbb{Z}_p, \qquad a(\xi) = 1, \qquad a'(\xi) \in \mathbb{Z}_p^\times, \qquad \beta(\xi) - n \in \mathbb{Z}_p^\times.$$

Proof. The polynomial

$$f(X) = (2X^2 + B)^8 + T^8 - 1$$

has derivative $32X(2X^2 + B)^7$. A common zero of f and its derivative would have either $X = 0$ or $2X^2 + B = 0$. These alternatives give, respectively,

$$B^8 + T^8 - 1 = 0, \qquad T^8 - 1 = 0,$$

both impossible because $B \geq 3$ and $T > 6$. Thus f is separable and $f(0) \neq 0$.

Let $G \in \mathbb{Z}[X]$ be a primitive integer multiple of the minimal polynomial of α. It is nonconstant and $G(0) \neq 0$. Exclude the primes dividing its leading coefficient or discriminant, the denominator of T, and the numerator and denominator of the norm of the nonzero number $a'(\alpha)$. Lemma 3.1 supplies infinitely many remaining primes where G has a simple root. Hensel's lemma lifts it to a root $\xi \in \mathbb{Z}_p$, thereby defining the required embedding. Its image $a'(\xi)$ is a unit; all finitely many denominator primes of the algebraic numbers in question may equivalently be included in the excluded set.

The rational number $n^8 + T^8 - 1$ is positive. Also exclude the primes dividing its numerator. If $\beta(\xi) \equiv n \pmod{p}$, then $a(\xi) = 1$ would give $n^8 + T^8 - 1 \equiv 0 \pmod{p}$, a contradiction. All the exclusions are finite. □

4 Local expansions of an arbitrary rational function

To allow dependence on the input as well as the two contact values, we use formal local coordinates. The integrality of all the coefficients is needed before specializing to rational primes.

Lemma 4.1. *Fix a root α of $a(X) - 1$, put $K = \mathbb{Q}(\alpha)$, and let $\Psi \in \mathbb{Q}(X, L)$ be nonzero. There are a finite set of rational primes S_0, an integer $C \geq 1$, and integers d_1, d_2 with the following property.*

Let $p \notin S_0$, let $\iota : K \hookrightarrow \mathbb{Q}_p$ be an embedding, and put $\xi = \iota(\alpha)$. Suppose $X, L \in \mathbb{Z}_p$, $X \equiv \xi \pmod{p}$, and

$$\mu = v_p(L + 1) > 0, \qquad \nu = v_p(a(X)L + 1) > C\mu.$$

Then $\Psi(X, L)$ is defined and nonzero, and

$$v_p(\Psi(X, L)) = d_1\mu + d_2\nu. \tag{7}$$

Proof. Write $\Psi = P/Q$ with nonzero $P, Q \in \mathbb{Q}[X, L]$. By inverting finitely many rational primes in $\mathcal{O}_K$, choose a ring A containing α, all coefficients of a, P, Q, and both $a'(\alpha)$ and its inverse. Thus

$$a(\alpha + h) - 1 = c_1 h + \cdots + c_{16} h^{16}, \qquad c_j \in A, \qquad c_1 \in A^\times.$$

There is a unique formal inverse

$$\mathcal{H}(z) \in zA[[z]], \qquad a(\alpha + \mathcal{H}(z)) - 1 = z.$$

Indeed, the first coefficient is c_1^{-1}, and at every subsequent degree the new coefficient is obtained from earlier ones by ring operations and division by c_1. Consequently no additional denominator primes occur.

Set

$$u = L + 1, \qquad v = a(X)L + 1, \qquad L = u - 1, \qquad X = \alpha + \mathcal{H}\left(\frac{u - v}{1 - u}\right). \tag{8}$$

These are inverse changes of formal coordinates at $(X, L) = (\alpha, -1)$; in particular $a(X) - 1 = (u - v)/(1 - u)$. The substitution sends P, Q to nonzero series $\widetilde{P}, \widetilde{Q} \in A[[u, v]]$. Nonvanishing follows because the two coordinate substitutions are inverse isomorphisms of the completed local rings.

For a nonzero series $F = \sum c_{ij} u^i v^j$, let j_F be the smallest j with a nonzero coefficient, and then let i_F be the smallest i occurring with $j = j_F$. For positive weights μ, ν satisfying $\nu > i_F \mu$, the term $c_{i_F j_F} u^{i_F} v^{j_F}$ is uniquely of smallest weight. Terms with the same j have larger i, while any term with larger j has excess weight at least $\nu - i_F \mu > 0$.

Apply this observation to $\widetilde{P}, \widetilde{Q}$ and take $C = 1 + \max(i_P, i_Q)$. Enlarge the excluded set of rational primes so that the two selected nonzero algebraic coefficients are units at every remaining embedding into $\mathbb{Q}_p$. This requires only finitely many primes: after the coefficients are integral, exclude the numerator primes of their nonzero field norms. All coefficients of the two series are now p-integral, and their selected coefficients are units.

The series converge whenever $\mu, \nu > 0$, since the valuation of a term tends to infinity with its total degree. For $\nu > C\mu$, unique minimality therefore gives

$$v_p(\widetilde{P}) = i_P \mu + j_P \nu, \qquad v_p(\widetilde{Q}) = i_Q \mu + j_Q \nu.$$

The actual X is on this formal inverse branch: its residue is ξ, and the unit derivative gives a unique solution in that residue class to $a(X) - 1 = (u - v)/(1 - u)$. Thus these are the valuations of $P(X, L), Q(X, L)$. Both are nonzero, and (7) follows with

$$d_1 = i_P - i_Q, \qquad d_2 = j_P - j_Q.$$

□

5 Realizing the two contact valuations

At the common contact -1, direct differentiation gives

$$R'(-1) = -\frac{3}{5120}. \tag{9}$$

Hence, for $p \notin \{2, 3, 5\}$ and $z \equiv -1 \pmod{p}$,

$$v_p(R(z)) = v_p(z + 1). \tag{10}$$

Lemma 5.1. *Let p be odd and let $\lambda_* \in \mathbb{Z}_p$ satisfy $\lambda_* \equiv -1 \pmod p$. For every positive integer M there exist integers s, t such that*

$$s^2 + t^2 \equiv \lambda_* \pmod{p^M}.$$

The integer $s^2 + t^2$ can be chosen arbitrarily large.

Proof. The sets of residues u^2 and $-1 - v^2$ modulo p both have cardinality $(p+1)/2$, so they intersect. Thus $u^2 + v^2 = -1$ has a solution over $\mathbb{F}_p$, and its two coordinates cannot both be zero. Interchange them if necessary so that $u \neq 0$. Fix an integer lift t of v. Hensel's lemma, applied to $S^2 + t^2 - \lambda_*$, gives a root $s_* \in \mathbb{Z}_p^\times$. Any integer $s \equiv s_* \pmod{p^M}$ gives the required congruence. Adding arbitrary integer multiples of p^M to s makes the norm arbitrarily large. □

Lemma 5.2. *For each embedding furnished by Lemma 3.2 outside $\{2, 3, 5\}$, with $\xi = \iota(\alpha)$, and for every pair of positive integers μ, ν, there are X, λ satisfying (3), (4), $X \equiv \xi \pmod p$, and $v_p(\beta(X) - n) = 0$. The quantities λ and $|X|$ can be chosen arbitrarily large after fixing μ, ν.*

Proof. Choose ξ as in Lemma 3.2, and put

$$\lambda_* = -1 + p^\mu, \qquad \tau_* = -1 + p^\nu, \qquad a_* = \tau_*/\lambda_*.$$

Both λ_* and τ_* are units, and $a_* \equiv 1 \pmod p$. Since $a'(\xi)$ is a unit, Hensel's lemma gives $X_* \equiv \xi \pmod p$ with $a(X_*) = a_*$. Equivalently, the two-variable map

$$(X, \lambda) \longmapsto (R(\lambda), R(a(X)\lambda))$$

has, at $(\xi, -1)$, Jacobian matrix

$$\begin{pmatrix} 0 & R'(-1) \\ -a'(\xi)R'(-1) & R'(-1) \end{pmatrix}, \qquad \det = R'(-1)^2 a'(\xi) \in \mathbb{Z}_p^\times.$$

Choose $M > \max(\mu, \nu)$. By Lemma 5.1, take $\lambda = s^2 + t^2 > 1$ congruent to λ_* modulo p^M. The Chinese remainder theorem gives an odd integer q such that

$$q \equiv 2X_* \pmod{p^M}.$$

Set $X = q/2$. Then $X \in \frac{1}{2}\mathbb{Z} \setminus \mathbb{Z}$ and $X \equiv X_* \pmod{p^M}$. The coefficients of a are p-integral, so

$$\lambda \equiv -1 + p^\mu \pmod{p^M}, \qquad a(X)\lambda \equiv -1 + p^\nu \pmod{p^M}.$$

Their respective contact depths are exactly μ and ν; (10) proves (4). Moreover $X \equiv \xi \pmod p$, so $\beta(X) - n$ remains a unit. Arbitrarily large lifts of q modulo $2p^M$, independently of the arbitrarily large norms in Lemma 5.1, give the last assertion. □

Proof of Theorem 1.1. Choose a root α of $a(X) - 1$ and apply Lemma 4.1. The linear map

$$(\bar\mu, \bar\nu) \longmapsto d_1\bar\mu + d_2\bar\nu$$

from $\mathbb{F}_2^2$ to $\mathbb{F}_2$ has a nonzero kernel vector ϵ. Add the finite exceptional set of that lemma and $\{2, 3, 5\}$ to S. Lemma 3.2 gives infinitely many remaining primes and embeddings. Fix one and any positive depths satisfying $\nu > C\mu$ and $(\mu, \nu) \equiv \epsilon \pmod 2$. Lemma 5.2 realizes these depths with all the required global parameter conditions, in the residue class of the selected root. Lemma 4.1 then gives a defined nonzero value of Ψ of even valuation.

The chosen parity class meets the cone with both coordinates arbitrarily large. For example, one may take

$$\mu = 2k + \epsilon_1, \qquad \nu = 2(C+1)\mu + \epsilon_2, \qquad k \geq 1.$$

The final assertion concerning the real sizes of the parameters follows from Lemma 5.2. □

Corollary 5.3. *The conclusion of Theorem 1.1 holds with* $\Psi(X,\lambda) = \Phi(R(\lambda), R(a(X)\lambda))$ *for every nonzero* $\Phi \in \mathbb{Q}(Z_0, Z_1)$.

Proof. At $(X,\lambda) = (\alpha, -1)$, the contact map has Jacobian determinant $R'(-1)^2 a'(\alpha) \neq 0$. Equivalently, its components are formal local coordinates, by (8) and the nonzero linear term of $R(-1+u)$. Hence it is dominant and its pullback on rational function fields is injective. The displayed Ψ is therefore nonzero, so the theorem applies. □

Corollary 5.4. *All conclusions of Theorem 1.1 remain valid with the additional requirement*

$$\lambda = 1 + s^2, \qquad s \in \mathbb{Z} \setminus \{0\}.$$

Thus $\lambda - 1$ *may be required to be an integer square, while* λ *is still a sum of two integer squares.*

Proof. In the proof of the theorem, replace $K = \mathbb{Q}(\alpha)$ by the finite extension $K' = \mathbb{Q}(\alpha, \sqrt{-2})$. The formal coefficients and their integrality conditions are unchanged after this extension.

Choose a nonzero primitive algebraic integer η for $K'/\mathbb{Q}$, with minimal polynomial $G \in \mathbb{Z}[X]$. Such an η is obtained by multiplying any primitive element by a suitable nonzero integer. The polynomial G is monic and separable, and $G(0) \neq 0$. Outside any finite set, Lemma 3.1 gives infinitely many primes at which G has a simple root. Hensel lifting gives embeddings $K' \hookrightarrow \mathbb{Q}_p$. Enlarge the finite excluded set by the denominator primes in expressions for α and $\sqrt{-2}$ as rational polynomials in η, as well as the earlier exceptional primes. The images of all required coefficients are then integral or units exactly as in the proof of the theorem.

In particular, the image ζ of $\sqrt{-2}$ is a p-adic unit with $\zeta^2 = -2$. For the desired local slope $\lambda_* = -1 + p^\mu$, Hensel's lemma gives a unit s_*, congruent to ζ modulo p, such that

$$s_*^2 = \lambda_* - 1.$$

For $M > \max(\mu, \nu)$, choose arbitrarily large integers $s \equiv s_* \pmod{p^M}$, and put $\lambda = 1 + s^2$. This realizes the same slope jet as in Lemma 5.2. The construction of the half-integral input and every subsequent valuation calculation are unchanged. □

6 Exact examples and scope

Theorem 1.1 has the following exact examples. Take

$$p = 76213, \qquad B = 3, \qquad n = 1, \qquad T = 57/4.$$

The last value is the trace of the abscissa of the point P in [1]. The residue $\xi = 5598$ is a simple root of $a(X) - 1$ modulo p:

$$\beta(\xi) \equiv 28125, \qquad a'(\xi) \equiv 22560 \pmod{p}.$$

For the product $\Phi(Z_0, Z_1) = Z_0 Z_1$, set

$$X = \frac{6603791433}{2}, \qquad \lambda = 3821558707^2 + 3^2.$$

Then the valuations of $R(\lambda)$, $R(a(X)\lambda)$, their product, and $\beta(X) - 1$, in that order, are

$$(1, 1, 2, 0).$$

For the harmonic combination $\Phi(Z_0, Z_1) = Z_0 Z_1/(Z_0 + Z_1)$, set

$$X = \frac{24884609587871}{2}, \qquad \lambda = 304951751852576^2 + 3^2.$$

The corresponding valuations are

$$(1, 2, 2, 0).$$

The supplied exact-arithmetic checker verifies primality, the root and derivative residues, the norm formulas, and all four valuations in both examples. These finite checks illustrate the theorem; the universal assertion is proved in the preceding sections. The choice $B = 3$ in the examples is a coefficient fixture, not a claim that this shift certifies the thirteen-variable definition. The theorem applies separately to every fixed odd $B \geq 3$.

The obstruction is restricted to demanding that the odd valuation of one fixed nonzero rational function detect the union of the two odd contact valuations uniformly. It includes all rational combinations of the two contact values and also permits explicit rational dependence on the input, the scale, and the slope. It persists under the scale and positivity conditions above.

It does not assert that the constructed nonintegral parameters satisfy the elliptic tests or the pole equations of the full Diophantine source. It does not exclude a different height estimate, a construction using additional arithmetic conditions, functions depending on further independent elliptic coordinates or tests, or a definition with eleven or fewer variables. No uniform quantitative bound for the primes in the theorem is asserted.

References

[1] *Thirteen Variables from a Weak abc Inequality*, 9 November 2026, project paper, conditional13_20261109_v16.tex.